Supplement: GeoCert: Certified Expression Intervals under Segmentation Uncertainty in Spatial Transcriptomics

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Sections 1–5 give the continuous realization proofs, atlas construction and count algorithms. The cut representation handles general assignment objectives and coupled boundaries; the count chain solves hard budgets for single-boundary point transcripts. Both use the same minimum-displacement realization; the disjoint-ribbon atlas converts this cost to physical changed-label area. Sections 6–8 report the original crops and width study, Section 9 tests correctness and runtime, and Section 10 gives the biological donor replication and paired Proseg analysis. Complete numerical tables accompany the figures so that each comparison can be reconstructed.

1 Continuous realization of assignments

Use coordinate t along an interface and x across it. Let m ordered boundary graphs $f_a : I \to \mathbb{R}$, $a = 0, \dots, m-1$, divide a compact corridor into $m+1$ cells. Each boundary has a constant slope limit $s_a > 0$, a constant band $[\underline{b}_a, \overline{b}_a]$, and fixed endpoint values. Adjacent boundaries satisfy

$$0 < d_a^{\min} \le f_{a+1}(t) - f_a(t) \le d_a^{\max}. \tag{1}$$

The upper bound may be absent. A rational piecewise-affine reference f^0 satisfies these geometric constraints. Exterior rails strictly beyond the bands close the outer cells.

Site i is a rectangle $[x_i^-, x_i^+] \times I_i$ in group $g(i)$; a point is a zero-size rectangle. All sites in a group share cell label c_g. Threshold bits $b_{ga} = \mathbf{1}[c_g \le a]$ satisfy $b_{ga} \le b_{g,a+1}$. For clearance $\rho > 0$, requiring the entire site to lie on its assigned side gives

$$b_{g(i),a} = 1 \Rightarrow f_a(t) \ge \ell_i = x_i^+ + \rho, \qquad b_{g(i),a} = 0 \Rightarrow f_a(t) \le u_i = x_i^- - \rho \quad (t \in I_i). \tag{2}$$

Fixed structural regions have pinned labels and no molecular reward. Nominal RNA locations use point sites. Universal-location regions use (2); they can make the source assignment inadmissible even though f^0 satisfies the geometry.

For a complete assignment c, let $\mathcal{F}(c)$ contain its realizing boundaries and define

$$D^\star(c) = \min_{f \in \mathcal{F}(c)} \sum_a \int_I |f_a(t) - f_a^0(t)|\, dt, \qquad \min \varnothing = +\infty. \tag{3}$$

This is summed interface displacement in the declared coordinates. In a general coupled family, swept regions may overlap, so this sum need not equal the physical area whose label changes. The disjoint two-label ribbon construction in Section 2 gives that equality after the chart Jacobian is applied. For arbitrary unary molecular costs $C_g(c_g)$ and $\lambda \ge 0$, the penalized objective is $\lambda D^\star(c) + \sum_g C_g(c_g)$; a hard budget is a different problem.

1.1 Propagation and exact feasibility

Give the boundary-index graph an edge $a \to a+1$ of weight $d_a^{\min}$ and, when finite, a reverse edge of weight $-d_a^{\max}$. Let Δ_{ab} be maximum path weight, with $\Delta_{aa} = 0$ and $-\infty$ for no path. Source feasibility excludes

positive cycles. Define

$$K_{ab}(r) = \max_v \{\Delta_{av} + \Delta_{vb} - s_v r\}, \qquad r \geq 0. \tag{4}$$

Lemma 1 (Heterogeneous propagation)**.** *Every feasible family obeys $f_b(y) \geq f_a(x) + K_{ab}(|x-y|)$. This is the strongest inequality obtained by finite compositions of the stated slope and same-coordinate difference constraints.*

Proof. For any intermediate boundary v, transfer from a to v at x, travel on v to y, and transfer to b. This gives each term of (4). Conversely choose a boundary of smallest slope rate among the longitudinal portions of a composite path. Its vertical portions before and after that visit are bounded by Δ_{av}, Δ_{vb}. Total absolute longitudinal travel is at least $|x-y|$, at rates at least s_v. Hence the path cannot improve that term. The zero-distance case follows from transitive closure. □

The kernels are nonincreasing, $K_{ab}(0) = \Delta_{ab}$, and $K_{ab}(r_1 + r_2) \geq K_{av}(r_1) + K_{vb}(r_2)$. A term with rate greater than the destination rate s_b is dominated by its direct-b term; the analogous source statement holds.

A lower requirement (a, I_i, ℓ_i) is forbidden if a kernel term (d, s) to any b satisfies $\ell_i + d > \overline{b}_b$, or $\ell_i + d > f_b^0(e) + s\,\mathrm{dist}(e, I_i)$ at either endpoint e. An upper requirement (a, I_i, u_i) is forbidden if a term of K_{ba} satisfies $u_i - d < \underline{b}_b$, or $f_b^0(e) > u_i - d + s\,\mathrm{dist}(e, I_i)$. Add nesting and all fixed-label bits. Finally add

$$b_{g(i),a} \Rightarrow b_{g(j),b} \quad \text{whenever } \ell_i + K_{ab}(\mathrm{dist}(I_i, I_j)) > u_j. \tag{5}$$

These are finite unary restrictions and implications; a contradiction means an empty geometric family.

Theorem 2 (Feasible assignments)**.** *The finite restrictions characterize exactly the complete assignments realizable by the continuous family.*

Proof. Necessity follows from Lemma 1; minimum interval distance gives the largest propagated lower requirement. For sufficiency, propagate all active lower requirements, bands and fixed endpoint values into $L_b(t)$, and upper requirements into $U_b(t)$. A lower term is $\ell_i + K_{ab}(\mathrm{dist}(t, I_i))$; an upper term from a is $u_i - K_{ba}(\mathrm{dist}(t, I_i))$. For a lower requirement on a, I_i and an upper requirement on c, I_j, put $r_i = \mathrm{dist}(t, I_i)$ and $r_j = \mathrm{dist}(t, I_j)$. Then

$$K_{ab}(r_i) + K_{bc}(r_j) \leq K_{ac}(r_i + r_j) \leq K_{ac}(\mathrm{dist}(I_i, I_j)) \leq u_j - \ell_i.$$

The first inequality is kernel composition, the second uses monotonicity and the interval triangle inequality, and the last is (5). Every lower term therefore lies below every upper term. Unary restrictions handle fixed data, so $L_b \leq U_b$. For a directed gap edge $a \to b$ of weight d, closure gives $K_{vb} \geq K_{va} + d$ and $K_{av} \geq d + K_{bv}$. Thus $L_b \geq L_a + d$ and $U_a \leq U_b - d$. Applying this to reverse edges enforces maximum gaps as well. Dominated rates preserve individual slope limits. Fixed data enforce bands and endpoint equality. Both envelope families therefore satisfy the geometry and realize the selected requirements. Nesting decodes one label per group. □

1.2 An exact cost function before molecular objectives

Theorem 3 (Assignment-wise cost representation)**.** *For rational input, a polynomial-size directed graph with assignment bits $b(c)$, helpers y, rational capacities, hard implications and known constant κ satisfies*

$$\min_y \mathrm{Cut}_{G_D}(b(c), y) + \kappa = D^\star(c) \tag{6}$$

for every feasible c. Hard-feasible cuts exist exactly for feasible assignments. For arbitrary rational unary molecular costs and rational $\lambda \geq 0$, a cut either detects infeasibility or returns an optimum and an attaining rational piecewise-affine boundary.

Proof. The family of lower difference inequalities is closed under componentwise minimum and maximum; those operations also preserve each scalar slope bound, band and endpoint. Since L, U, f^0 obey the geometry,

$$f_b^*(t) = \min\{U_b(t), \max\{L_b(t), f_b^0(t)\}\} \tag{7}$$

is jointly feasible and satisfies the selected requirements. Every realization has $L_b(t) \le f_b(t) \le U_b(t)$. The clamp is the nearest permitted value to $f_b^0(t)$ at every coordinate, proving simultaneous minimum displacement and (3).

An upward displacement is the maximum of nonnegative functions $[\ell_i + d - s\,\mathrm{dist}(t, I_i) - f_b^0(t)]_+$ activated by bits; downward displacements use complementary bits. Include structural requirements and source-inconsistent RNA requirements. Upward and downward contributions cannot both be positive on a feasible assignment. Split the domain at their affine-piece endpoints, zeros, source knots and pair crossings. On one slab let $h_1 \ge \cdots \ge h_k > 0$, $h_{k+1} = 0$, with S_j the labels in the first j activations. Then

$$\max_{i \text{ active}} h_i = \sum_{j=1}^{k} (h_j - h_{j+1}) \mathbf{1}[S_j \text{ contains an active bit}]. \tag{8}$$

Integrate the nonnegative affine differences exactly. Take bit one to mean the source side of a cut. For $w\,\mathrm{OR}_{i \in S} b_i$, add hard arcs $b_i \to y$ and a capacity-w arc $y \to$ sink. For $w\,\mathrm{OR}_{i \in S}(1 - b_i)$, use a capacity-$w$ arc source $\to y$ and hard arcs $y \to b_i$ [1, 2]. Minimizing the helper gives exactly the indicated cost in both cases. Repeated labels are included once. Fixed terms are retained in κ or forced helpers. Thus minimizing over helpers gives the exact integrated clamp for each fixed $b(c)$, proving (6). Expand any unary cost as $C_g(m) + \sum_{a<m}[C_g(a) - C_g(a+1)]b_{ga}$ and shift negative terms by a known constant. Rational scaling yields integer capacities. After molecular costs and λ are chosen, each numerical hard capacity is set above the resulting total finite capacity. The fixed object is the geometric representation and its logical implications, rather than one numerical big-M for every possible future objective. A cut through a hard arc distinguishes infeasibility; (7) reconstructs the boundary. □

Graph size versus construction work. Let N count all merged RNA and structural rectangles, G groups, and $M_b = O(Nm^2)$ propagated terms on boundary b. For one polarity the untruncated terms are $g_i(t) = \alpha_i - s_i\,\mathrm{dist}(t, I_i)$. Their common reference changes positivity, but not relative cone order. Each pair changes order only constantly many times. An adjacent swap changes one full prefix; starting with M_b prefixes yields $O(M_b^2 + M_b)$ distinct activation sets. Simultaneous events may be resolved into adjacent swaps, repeated labels only merge sets, and zero-weight predecessor prefixes support constant-arc helpers. Consequently

$$|V| = O\left(Gm + \sum_b (M_b^2 + M_b)\right), \quad |E| = O\left(Gm + N^2m^2 + \sum_b (M_b^2 + M_b)\right). \tag{9}$$

The auxiliary contribution is $O(N^2m^5)$. Source knots still affect integration, coefficient lengths and reconstruction. Exact intersections, integrals and denominator scaling have polynomial bit complexity. An event integrator updates affected adjacent layers rather than rescanning every layer; on a J-knot reference its work is $O((M^2 + MJ + J + 1)\log(M + J + 2))$ rational-arithmetic and bitset operations, given that reference. Queueing adjacent inversions handles multiple crossings; unchanged layer integrals accumulate until the next change. These bounds describe graph construction, with rational operand costs accounted for separately. Arrangement sweeps are established tools [3].

2 Physical atlas and exact reference agreement

The reference is the original labelled raster interpreted as a union of closed pixel squares with disjoint interiors. A positive cell label is eligible when its pixel set is connected and does not meet the crop boundary. Background label zero may participate as the other side of an interface; it is not an additional biological cell. Ineligible cells, crop boundaries and unselected portions of interfaces remain fixed.

For each adjacent eligible label pair, trace shared pixel edges into paths. Split paths at vertices where continuation is ambiguous and into coordinate-monotone runs. Divide long runs at a maximum of 128 unit edges, trim one edge from each end, and retain at least four remaining edges. Rank candidates by length and

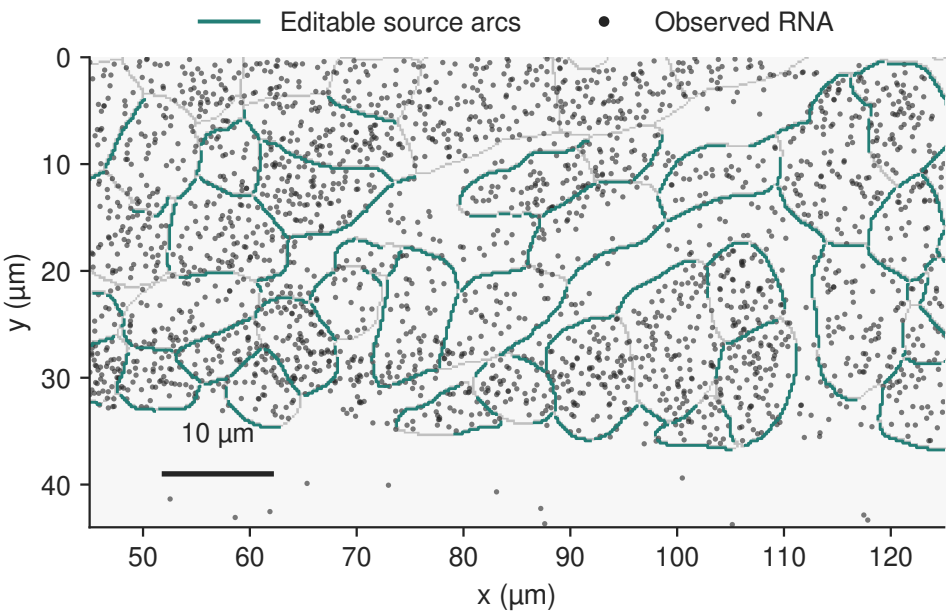


Figure 1: Reference labels and observed RNAs in a lung-crop excerpt. Lines mark editable source arcs; the scale bar is 10 μm. Ribbons belong to one disjoint atlas. Their endpoint seams, junction neighborhoods and uncovered interfaces stay fixed.

then by a deterministic label/coordinate ordering. Try ribbon halfwidths $4, 2, 1, 1/2$ pixels in that order. No gene identity, transcript count, count interval or optimization outcome determines this ranking.

Let $s_x, s_y \in \{-1, 1\}$ orient a monotone run. Pixel-edge coordinates obey

$$t = s_x x + s_y y, \qquad u = s_x x - s_y y.$$

The run is exactly $u = r(t)$, with slopes $+1$ or -1 between its recorded raster vertices. Put $v = u - r(t)$. The inverse physical chart, up to its stored origin, is

$$\Phi(t, v) = \frac{p}{2}\big(s_x(t + r(t) + v),\ s_y(t - r(t) - v)\big), \qquad |\det D\Phi| = \frac{p^2}{2}.$$

The pixel size p is read from the source coordinate transformation and retained as an exact rational value. An accepted ribbon has $v \in [-B, B]$, allowed displacement $|h(t)| \le 3B/4$, $|h(t) - h(t')| \le |t - t'|$, and zero endpoint displacement. The RNA clearance is $\rho = 10^{-6}/p$ in chart coordinates; it separates the two admissible ownership sides numerically. Together these settings define the geometric sensitivity family.

2.1 Acceptance and nuclear constraints

Every piece of the left/right half-ribbon is clipped against intersecting native pixels using rational polygon arithmetic. Positive-area intersections must have the declared left/right source labels. Pairwise ribbon interiors must be disjoint. A geometric spatial index selects candidate pairs for the final rational intersection test. RNA coordinates are transformed through the exact chart and their nominal side must agree with the original source mask. A molecule may occur in at most one accepted patch.

A nuclear-mask pixel is labelled by the source *cell* mask at that pixel, rather than by its unrelated nucleus identifier. Clip its square to the ribbon, transform its fragments into (t, v) coordinates, and split at reference-curve knots. Over each projected interval, the maximum required left-side value or minimum right-side value supplies a conservative constant bound. The resulting conservative bound preserves the

label of every intersecting nuclear fragment. Structural constraints have zero additional clearance. Encoding structural regions as shifted point/interval constraints cancels the engine's RNA clearance exactly, without adding molecular counts.

An initial extraction pilot omitted nuclei. The final procedure added nuclear constraints and cell–background interfaces before recomputing counts and transferring the fixed rules to the breast crops. The provenance record preserves both stages.

2.2 Global homeomorphism

For fixed longitudinal coordinate and any admissible h, define

$$H_h(t,v) = \begin{cases} -B + (v+B)(h(t)+B)/B, & -B \le v \le 0, \\ h(t) + v(B-h(t))/B, & 0 \le v \le B. \end{cases}$$

Its transverse slopes are strictly positive because $|h| < B$. Both formulas agree at zero, map zero to h, and fix $-B, B$. Its inverse is continuous, and $h = 0$ at longitudinal endpoints makes the whole boundary trace fixed. Conjugating $(t,v) \mapsto (t, H_h(t,v))$ through Φ gives a homeomorphism of the closed physical ribbon fixing its boundary. It maps the two source sides to the requested alternatives.

Extend each map by the identity outside its ribbon. Disjoint interiors and fixed traces make the maps agree on intersections. A finite union is therefore a global continuous bijection with the corresponding piecewise inverse. The reference cell topology, connectivity, holes and junction incidences are preserved.

Changed-label area inside one two-label ribbon is exactly

$$A(h) = \frac{p^2}{2} \int_I |h(t)|\, dt.$$

Interiors do not overlap, so changed-label area in the atlas is the sum of these costs, counted once. At source knots the chart is piecewise affine; the finite kink set has zero area. The constant Jacobian makes this conversion exact; other coordinate charts require their corresponding area factor.

3 Exact assignment-cost function and graph construction

In an ordered corridor, each boundary has a slope bound, band and endpoint values; adjacent boundaries have constant minimum and optional maximum separations. All finite rational data are binary encoded. The source satisfies the geometric constraints. RNA-location rectangles and pinned structural regions induce lower or upper interval requirements depending on their discrete owner thresholds.

The maximum-path closure Δ of same-coordinate separation inequalities yields

$$K_{ab}(r) = \max_c \{\Delta_{ac} + \Delta_{cb} - s_c r\}.$$

Every finite sequence of permitted longitudinal and cross-boundary steps is bounded by such a term: choose the least slope on that path, bound its total travel below by the endpoint distance, and its remaining directed weights above by the corresponding closure weights. This gives the complete propagated lower/upper restrictions, including both directions of each finite maximum gap.

The selected lower family L, upper family U, and reference f^0 obey the same geometric difference inequalities. The finite conflict tests imply $L \le U$. Minimum and maximum preserve these inequalities and each scalar slope bound, so

$$f_b^*(t) = \min\{U_b(t), \max\{L_b(t), f_b^0(t)\}\}$$

is a jointly feasible family and is pointwise nearest to the reference. This simultaneous attainment establishes the minimum $D^*(c)$ for each complete assignment.

3.1 Integrated layers and the helper bound

In one direction, source-relative excursions are maxima of active flat-top cones. On an analytic interval with ordered positive heights $h_1 \geq \cdots \geq h_q > 0$, set $h_{q+1} = 0$. Their activated maximum equals

$$\sum_{j=1}^{q} (h_j - h_{j+1})\, \mathbf{1}[\text{at least one activation among the first } j \text{ occurs}].$$

Integrate the affine differences exactly. Repeated cones carrying the same molecular bit use that bit once; complementary activations encode the opposite direction. Fixed-positive constraints contribute a mandatory constant or forced helper. Standard implication/OR gadgets then preserve the exact integral for every hard-feasible owner vector. Finite hard capacities exceed the total finite capacity for the requested objective.

For M untruncated flat-top cones, a pair changes order only constantly many times. Starting with M prefixes, each adjacent swap introduces at most one additional prefix. Thus the union contains $O(M^2 + M)$ distinct sets. A common reference changes positivity, not cone order. Resolve simultaneous events into adjacent swaps and retain a deterministic tie rule on coincidence intervals. Mapping cone indices to repeated activation labels cannot enlarge the family. A set can use a predecessor prefix plus its last newly added label; retaining zero-weight predecessors keeps the arc count proportional to the number of sets.

If $M_b = O(Nm^2)$ includes all propagated molecular and structural constraints, the graph has

$$|V| = O\left(Gm + \sum_b (M_b^2 + M_b)\right), \quad |E| = O\left(Gm + N^2m^2 + \sum_b (M_b^2 + M_b)\right).$$

The auxiliary contribution is $O(N^2m^5)$. Source knots, intersections and coefficient denominator lengths still affect construction and encoding. Event-driven accumulation can integrate a direction's coefficients in $O((M^2 + MJ + J + 1)\log(M + J + 2))$ rational-arithmetic and bitset operations for a J-knot reference. This integration-stage statement is distinct from the final graph-size bound and from total bit complexity. The critical-height algorithm below uses this canonical realization directly to recover every count-specific minimum, including states unavailable to a penalty sweep.

4 Critical-height count algorithm

4.1 Input contract and common mesh

The count chain specializes the canonical realization to one boundary and one point per free RNA group, with fixed structural regions, a rational piecewise-affine source, slope bounds, bands and endpoints. A free group spanning several locations or an interval would require ownership memory across nodes and lies outside this input contract. Fixed groups may contain many structural rectangles because their labels are known. Group scores are nonnegative integers for each owner, with an integer fixed offset. Atlas, source-observation and saved exact-geometry identities are checked before certification; the count chain checks the raw point-input contract before compilation.

Collect all affine pieces that any canonical realization can select: lower and upper cones for every possible point owner, fixed constraints, bands, endpoints and source. Include all piece endpoints, observation locations and same-boundary intersections in a common mesh. Omit crossings outside the forced envelope only when they cannot be a canonical switch. At every node, candidate heights are function values inside the forced range and outside every observation's clearance gap.

For any complete feasible assignment, its canonical minimizer selects affine pieces between mesh events and crosses the source only at events. Its nodal trapezoidal area is exactly its continuous area. Conversely a feasible candidate path interpolates into a feasible boundary, and its trapezoid upper-bounds that interpolant's area. Retaining the canonical minimizer proves equality of minimum area at *each count*, not equality for every interpolant.

4.2 Recurrence, sliding minima and reconstruction

Let V_j be the sorted heights at time t_j, $z_j(v)$ the score of points observed at that time, and

$$a_j(v) = w_j|v - f^0(t_j)|, \qquad w_j = \tfrac{1}{2}[(t_j - t_{j-1}) + (t_{j+1} - t_j)],$$

with missing endpoint intervals omitted. Each point contributes once. Pinned-group scores enter the initial offset. The least prefix cost satisfies

$$F_j(v, q) = a_j(v) + \min_{\substack{u \in V_{j-1} \\ |u-v| \le s(t_j - t_{j-1})}} F_{j-1}(u, q - z_j(v)).$$

Initialize with first-node cost and score. For each predecessor count, scan candidate heights in increasing order. Insert each permitted predecessor into a monotone deque, discard dominated costs at its back, and discard expired heights at its front. The front gives the least predecessor cost. Both ends of the height-compatibility interval move monotonically, so one node transition takes $O(H_{j-1} + H_j)$ work per count. Backpointers reconstruct a complete owner vector and boundary, which are checked against the original continuous constraints and exactly integrated area.

If the variable score range is W, dynamic-programming work is $O((W+1)\sum_j H_j)$ and full backpointer storage has the same order. Value-only rolling storage can be smaller. With P explicit affine pieces, a conservative mesh bound is $O(P^2 + P)$ events, at most $O(P)$ heights per event, and $O((W+1)P^3)$ dynamic-programming work. For explicit unit transcripts, W is at most the number of scored molecules. Binary-compressed multiplicities instead give a pseudo-polynomial dependence on W. Rational operand lengths, candidate preparation and output serialization remain separate costs.

The implementation raises a resource-limit outcome instead of marking an incomplete spectrum exact. The reference is geometrically feasible but need not realize every point's original owner at zero cost. A positive mandatory-area test exercises that distinction. Single-copy source-feasible queries use the cheaper implication-closure fast path; all multicopy queries in the end-to-end count-chain census use the algorithm above.

4.3 An explicit unit-molecule penalty gap

Use source zero on $[0, 30]$, slope one, band $[-4, 4]$ and clearance $\rho = 1/10$. At centers 5, 15 and 25 place respectively 2, 3 and 3 distinct point RNAs. Point j in a cluster has $(t, x) = (\text{center} + j/100,\ h - j/100 - \rho)$, with $h = 2, 3, 3$. The slope and clearance force a common owner within each cluster; their leading tents dominate, and clusters have disjoint positive supports. Full owner enumeration gives eight feasible vectors and spectrum

$$\{(0,0), (2,4), (3,9), (5,13), (6,18), (8,22)\}$$

(count, area). At area ten the largest attainable count is three. Every penalty bound is at least four: for $\mu \le 1/3$ the count-eight state gives $8 - 12\mu \ge 4$, and for $\mu \ge 1/3$ the count-two state gives $2 + 6\mu \ge 4$. At $1/3$ equality holds. All molecules are unit observations at distinct coordinates; no succinct bundle encoding is used. The count chain recovers the unsupported count-three outcome.

5 Composition, baselines and joint predicates

For a cell–gene pair with source count n_0, index *every* patch of the complete atlas by $i = 1, \ldots, r$. Let $S_i(d)$ be the minimum physical area of a complete feasible local assignment with count *change* d, taking $+\infty$ when no such assignment exists. Then the minimum area for absolute count q is

$$S(q) = \min_{d_1 + \cdots + d_r = q - n_0} \sum_{i=1}^{r} S_i(d_i), \qquad [L, U] = \left[\min_{S(q) \le \delta} q,\ \max_{S(q) \le \delta} q \right].$$

An empty budget-feasible count set gives an empty uncertainty set. Using count changes includes fixed exterior molecules exactly once through n_0. For every finite minimizing trace, choose an attaining local

boundary on each patch, including patches that contribute zero count change. The gluing theorem gives a compatible partition whose physical area is the sum of these local areas. Conversely, restricting any feasible partition to its patches gives local count changes summing to $q - n_0$, and each local area is at least $S_i(d_i)$. These two inequalities prove the identity without assuming a zero-cost source assignment.

A patch with no contribution to the queried cell and gene has only change zero, but its minimum area $m_i = S_i(0)$ may be positive. If J contains every patch that can change the count, the same composition can be written

$$\beta = \sum_{i \notin J} m_i, \qquad S(q) = \beta + \min_{\sum_{i \in J} d_i = q - n_0} \sum_{i \in J} S_i(d_i).$$

The offset includes nonincident patches as well as incident patches without the target gene. Their minimizing boundaries must remain part of the global witness. If any required patch is infeasible, the complete atlas is infeasible. If every omitted patch has a feasible zero-area source witness, then $\beta = 0$ and those patches may remain unchanged. The tissue experiments use this special case, verified for every saved patch in §9.5. Missing or unfinished local spectra cannot be treated as zero-cost states.

The outer baseline lets every individually affordable RNA choose a feasible owner independently. It can combine incompatible changes or overspend the budget; its interval is an outer bound. The additive inner baseline retains exact local feasibility, charges every changed RNA its singleton minimum (including collateral genes), and composes these costs. Under zero-cost source admissibility, this sum upper-bounds shared area, so budget feasibility is an inner approximation. The comparison therefore distinguishes over-permissive independent transfers from over-conservative additive costs.

For co-detection, retain a two-dimensional score and compose vector spectra. Marginal endpoints need not coexist. The tested predicate is $N_{g_1} > 0$ and $N_{g_2} > 0$, classified as always, never or variable over the budgeted spectrum. The resulting classification describes the stability of observed co-detection under the chosen family.

5.1 Finite unions of complete atlas families

All member families use one reference partition, one retained observation set with fixed cell and gene identifiers, and the same changed-label physical-area functional. A member specifies its entire atlas, including fixed seams, uncovered regions, slope and nuclear constraints. For $\mathcal{F}_\cup = \bigcup_{a=1}^{A} \mathcal{F}_a$, every permissible partition is a member of at least one complete family.

For a count query N, define $S_a(q) = \min\{D(f) : f \in \mathcal{F}_a,\ N(f) = q\}$, with $+\infty$ for absent states. Then

$$S_\cup(q) = \min_a S_a(q).$$

Indeed, any union realization lies in a member and costs at least that member's minimum for its count, proving the lower bound. A member attaining the least finite minimum supplies the reverse bound and its entire partition as a witness. Since the number of members is finite and finite member minima are attained, every finite union minimum is attained. The budget-feasible count set is the union of the member feasible-count sets. Its minimum and maximum are therefore the minimum lower endpoint and maximum upper endpoint of the nonempty members. All-empty members yield an empty union.

Store each count's selected member identity together with the member's complete attaining trace. Tied minima are resolved by a fixed member order. The implementation combines spectra, never overlapping ribbons from different members. Thus each witness satisfies the original gluing theorem. Different count endpoints can select different members, just as marginal endpoints within one atlas can select different partitions. A joint marker or complete-matrix query must select a member for the entire query. If every member excludes an assignment vector, their union excludes that vector as well.

The unified question census contains every cell–gene pair with a positive source count or an editable observation in any member. A member missing a query contributes the singleton source count at zero area only when its atlas establishes that the count is fixed and its source partition is feasible at zero cost. A missing expected spectrum or an unresolved computation is an error, rather than evidence of invariance. Source counts and observation identifiers are checked across members before union. These checks distinguish fixed-count completion from silent changes of the measured population.

The union census contains 3,766 lung and 25,446 breast questions, yielding 29,212 spectra and 496,604 interval rows on the 17-budget grid. At area $4\,\mu\mathrm{m}^2$, 2,905/3,766 lung counts (77.1%) and 17,160/25,446 breast counts (67.4%) are invariant. For comparison on the unchanged primary question census, the corresponding counts are 2,524/3,304 (76.4%) and 15,143/22,976 (65.9%). The first pair of fractions describes the complete union output; the second isolates interval changes for the original questions.

For example, lung crop 1 cell 51003 has one source *CAVIN2* transcript. The primary family keeps this count fixed, whereas the $0.5\,\mu$m support-halfwidth member attains count zero, matching the adapted Proseg count. Its complete atlas witness uses one changed patch with area approximately $0.001161\,\mu\mathrm{m}^2$; all other patches remain at their sources. Thus the union's gains include alternative patch selection as well as larger allowed movements.

Area (μm^2)	Lung invariant	Breast invariant	Lung widened	Breast widened
0	3,304	22,976	0	0
0.005	3,155	21,482	8	65
0.01	3,102	20,976	9	82
0.025	3,006	20,045	12	142
0.05	2,924	19,126	15	202
0.1	2,844	18,279	33	364
0.15	2,784	17,734	42	462
0.2	2,731	17,360	55	527
0.25	2,710	17,097	77	838
0.375	2,663	16,575	128	1,532
0.5	2,629	16,293	169	1,979
0.75	2,602	15,961	208	2,461
1	2,582	15,762	230	2,747
2	2,542	15,342	279	3,332
4	2,524	15,143	298	3,595
8	2,517	15,113	306	3,643
16	2,517	15,113	306	3,643

Table 1: Union certificates on the unchanged primary question census: 3,304 lung and 22,976 breast cell–gene pairs. Invariant counts have singleton union intervals. Widened counts have at least one endpoint strictly beyond the primary interval at the same budget.

After the member computations, merging requires $O(\sum_a K_a)$ rational comparisons for K_a stored count states. There is no product over member states. Including the primary family gives $\mathcal{F}_{\mathrm{primary}} \subseteq \mathcal{F}_{\cup}$, so its feasible counts and reference-count reach on the same pairs are preserved. A union may therefore widen intervals and reduce the number of certified singleton counts. Its reported invariance denominator and interval table are recomputed for the declared union census.

For the union comparison, form each member's independent-molecule outer interval at the same budget, then take the smallest lower endpoint and largest upper endpoint across members. This envelope contains the exact union interval. We count a question once if the outer width reaches at least two anywhere on the 17-budget grid, and once as tightened if an exact endpoint is strictly inside the envelope at any grid budget. All tightened questions satisfy that eligibility rule. On the full union census, 29/102 lung and 791/2,076 breast questions tighten; on the primary census alone, the corresponding values are 28/101 and 783/2,058. These denominators describe the union outer interval, rather than the number of movable transcripts in an individual patch.

Reference	Changed counts	Primary attainable	Union attainable
Lung: Proseg	2,798	185 (6.6%)	315 (11.3%)
Lung: nuclear	1,114	184 (16.5%)	299 (26.8%)
Breast: nuclear	10,965	1,773 (16.2%)	2,942 (26.8%)

Table 2: Reference count changes attainable without an area cap, on the same cell–gene pairs. The union chooses among five complete atlases and contains every primary-atlas state. Each count is tested separately; different counts may require different partitions.

6 Data, extraction and complete experimental denominators

The lung inputs are two 200-micrometre crops of a public Xenium lung-cancer specimen, containing 7,370 retained Q20 measured-gene RNA records. The atlas is extracted directly from the source masks with complete reference-cell containment; it replaces the overlapping corridors of the earlier implementation.

The breast inputs are the two previously acquired source crops in the public workflow artifact identified in the provenance record. The artifact's SHA-256 is `f53bee7d82a385a320c14a180049e66bf6f60963dda8d7a75574fda90c28febf`. It contains 81,182 retained RNA records and corresponding masks. The source is 10x Genomics' two-field-of-view breast format-test dataset, for which the vendor cautions against biological conclusions. We use these previously available crops to evaluate transfer of the fixed extraction and count-analysis rules.

	Lung 1	Lung 2	Breast 1	Breast 2
Retained RNA	5,847	1,523	43,366	37,816
Complete reference cells	79	74	253	214
Accepted patches	276	268	928	780
Editable RNA	627	118	4,789	3,473
RNA-bearing patches	190	72	760	628
Maximum RNA per patch	19	6	39	33
Editable reference edges	7,093	8,728	28,606	24,306
All incident reference edges	10,756	12,408	43,331	37,132

Table 3: Complete extraction census. Coverage denominators include incident edges with ineligible neighboring cells. Refer to the machine-readable coverage record for per-cell lengths and fractions.

The two lung crops have disjoint complete-cell identifiers, as do the two breast crops. Five eligible lung cells have no retained positive gene count and no editable relevant molecule; their rows are implicit zeros in the supplied matrix index. The interval CSV is sparse over every positive-source or editable cell–gene pair. Absent entries are zero only within the explicitly recorded panel/observed-gene universe, retained observations and atlas identity. Thus a sparse zero refers to retained observed counts.

All eight budgets $0, 0.05, 0.1, 0.25, 0.5, 1, 2, 4\,\mu\mathrm{m}^2$ are reported. There are 3,304 lung and 22,976 breast cell–gene spectra, yielding 26,432 and 183,808 unique interval rows (210,240 in total). At budget $4\,\mu\mathrm{m}^2$, 2,780 lung counts (84.1%) and 17,717 breast counts (77.1%) are invariant. One atlas per crop determines every interval and its compatible attaining boundary. Counts include RNA outside editable ribbons.

6.1 Budget-specific simple-baseline comparisons

In breast crop 1, cell 81's LDHB source count is 22. The minimum area attaining count 17 is exactly

$$\frac{19773488331902675966995674 29}{40850916262281216000000000000} \simeq 0.48404026497\,\mu\mathrm{m}^2.$$

The global trace changes three LDHB molecules through one patch and two through another, plus one collateral KRT8 molecule. The two patch areas are approximately 0.342327 and $0.141713\,\mu\mathrm{m}^2$; a third

Area budget	Lung variable	Breast variable	Breast outer tightened	Breast inner misses
0	0	0	0	0
0.05	366	3,684	84	11
0.1	429	4,414	80	25
0.25	524	5,259	61	29
0.5	524	5,259	3	3
1	524	5,259	0	0
2	524	5,259	0	0
4	524	5,259	0	0

Table 4: Counts out of 3,304 or 22,976 entries per budget. Lung outer improvements comprise four rows at 0.1; the two lung inner misses occur at 0.1 and 0.25. The differences concentrate at intermediate budgets, where shared area limits the combinations of affordable transfers.

incident LDHB patch remains unchanged. At budget 0.25 the exact interval is $[19, 23]$ versus marginal $[17, 23]$; at 0.5 it is $[17, 23]$ versus additive-inner $[18, 23]$. This example from the complete output shows how the two approximations fail in opposite directions.

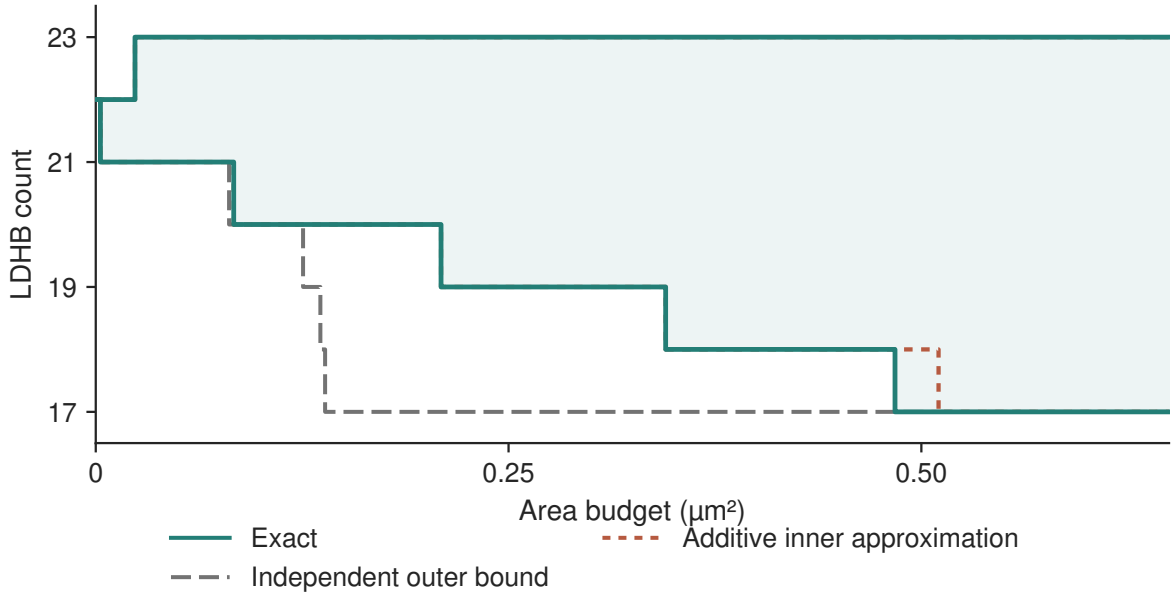


Figure 2: Exact and approximate LDHB intervals for breast cell 81. Each curve gives an upper or lower endpoint; shading marks the exact interval. Steps occur at the rational minimum areas in the saved spectra and singleton-transfer calculations. The independent outer bound permits jointly over-budget changes, while additive costs can miss an attainable endpoint. The displayed range contains all transitions through the $0.5\,\mu\text{m}^2$ example.

6.2 Question denominators and endpoint saturation

On the original eight budgets, 46 lung and 1,026 breast cell–gene questions have marginal width at least two for some budget. Exact geometry tightens four and 203 of these questions, affecting four of 153 lung cells and 139 of 467 breast cells. Relative to all 3,304 lung and 22,976 breast questions, the tightened fractions are 0.12% and 0.88%. Counting a question once for each affected budget gives four lung and 228 breast rows

with tighter outer bounds; the additive inner approximation misses two lung and 68 breast endpoints. A question is counted once in the question totals if at least one budget tightens an endpoint.

	Lung			Breast (Xenium)		
Area	Variable	Width sum	Endpoints changed	Variable	Width sum	Endpoints changed
0	0	0	0	0	0	0
0.05	366	380	366	3684	4279	3684
0.1	429	451	70	4414	5339	1028
0.25	524	572	120	5259	6754	1350
0.5	524	572	0	5259	6820	61
1	524	572	0	5259	6823	3
2	524	572	0	5259	6823	0
4	524	572	0	5259	6823	0

Table 5: Primary eight-budget occupancy, summed widths and number of cell–gene intervals whose endpoints change from the preceding displayed budget. A constant non-singleton total can coexist with further endpoint widening.

In the original atlas the largest support halfwidth is four chart units. The chart displacement has Euclidean norm $p|h|/\sqrt{2}$, hence its maximum allowed value is $3pB/(4\sqrt{2}) \simeq 0.4508\,\mu$m. This is the geometric displacement cap of the primary atlas. The lung and breast spectra attain their extreme endpoints by areas approximately 0.240525 and 0.573939. Constant non-singleton totals from area 0.25 therefore indicate occupancy saturation earlier than complete endpoint saturation.

7 Physical-width sensitivity with fixed selection rules

The prospective analysis record fixes required support halfwidths $r = 0.5, 1, 2, 3\,\mu$m, the original arc construction, run ordering, endpoint seams, slope one, nuclear constraints and clearance, and the area grid

$$0, .005, .01, .025, .05, .1, .15, .2, .25, .375, .5, .75, 1, 2, 4, 8, 16.$$

The protocol fixes these choices before width-dependent extraction and count calculations on the available inputs. Each run uses $B = \lfloor 4\sqrt{2}r/p \rfloor/4$ and accepts only that width. Rejected patches do not fall back to narrower widths. Longer arcs keep priority and all original coordinate tie rules. Consequently each width defines a separate fixed-seam atlas; the atlases need not be nested.

Every rejected candidate and accepted patch is retained in extraction summaries. Rejection can follow third labels, nuclei inconsistent with required clearance, chart geometry, intersections with earlier selected ribbons, or other declared acceptance conditions. All eligible source cells remain in count denominators, including those without editable patches. Each accepted patch is checked against exact raster-side intersections, all molecular source owners, structural constraints and disjoint interiors.

Support halfwidth	Displacement cap (μm)	Edge coverage (%) Lung	Breast	Lung Proseg reach (%)	Breast nuclear reach (%)
Adaptive	0.451	68.3	65.8	6.6	16.2
0.5	0.366	45.1	44.6	4.3	11.2
1	0.733	33.8	34.6	5.5	14.8
2	1.493	23.5	24.7	4.5	14.6
3	2.226	16.9	17.8	3.8	12.2

Table 6: Required physical support halfwidths are rounded down to quarter chart units. The cap is $3pB/(4\sqrt{2})$; actual feasibility also depends on other constraints. Reach is the fraction of changed reference counts attainable at any budget, with fixed denominators across families.

The last endpoint-saturation areas across Adaptive, 0.5, 1, 2 and 3 are, respectively, 0.240525, 0.154063, 1.497699, 3.090472 and 4.878380 for lung; and 0.573939, 0.376462, 1.281168, 5.095884 and 13.782726 for breast. Larger support permits larger movement, but fewer patches fit the same segmentation. The largest local gene multiplicity is 17; the largest patch contains 136 retained molecules. All eight count runs return complete exact spectra.

For like-for-like budget comparisons, the original adaptive-width spectra are also projected onto the 17-budget grid. This uses unchanged rational minima and individually affordable-transfer minima, and every original CSV row matches its earlier value exactly. On this finer grid, 12 of 46 lung and 361 of 1,026 breast questions tighten, versus the primary eight-grid results of 4 of 46 and 203 of 1,026. The finer grid resolves differences over shorter budget ranges in the same underlying spectra. New width columns use the same fine grid.

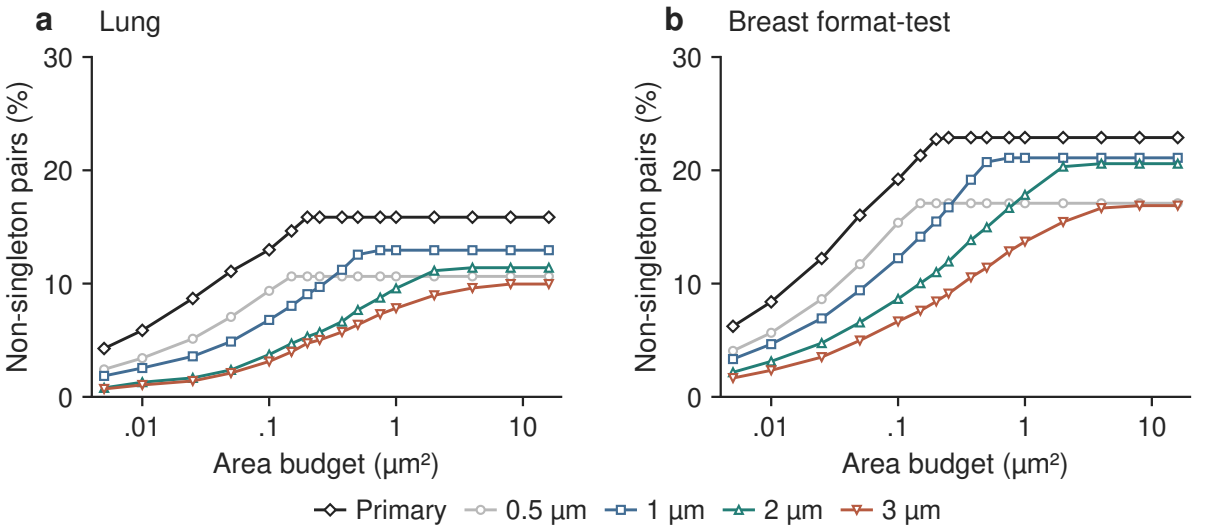


Figure 3: Fraction of non-singleton intervals in (a) lung and (b) breast format-test crops. Symbols mark positive tested budgets, with lines as visual guides. Each family's denominator includes all its positive-source or editable cell–gene pairs, so the curves summarize different geometric families. Budget zero is omitted from the logarithmic axis; every interval is a singleton there.

Area (μm^2)	Adaptive	0.5	1	2	3
0	0	0	0	0	0
0.005	141	78	61	27	24
0.01	194	110	84	44	36
0.025	287	165	118	57	48
0.05	366	227	161	81	72
0.1	429	301	224	127	107
0.15	484	342	265	160	136
0.2	524	342	299	181	162
0.25	524	342	320	194	172
0.375	524	342	370	226	196
0.5	524	342	414	261	218
0.75	524	342	427	298	250
1	524	342	427	326	268
2	524	342	427	379	307
4	524	342	427	388	329
8	524	342	427	388	341
16	524	342	427	388	341

Table 7: Lung: non-singleton intervals on the common 17-budget grid. Column labels after Adaptive are required physical support halfwidths in micrometres. The numbers of question entries for the five families are 3304, 3216, 3298, 3403, 3424.

Area (μm^2)	Adaptive	0.5	1	2	3
0	0/0	0/0	0/0	0/0	0/0
0.005	0/0	0/0	0/0	0/0	0/0
0.01	1/3	1/2	0/2	0/0	0/0
0.025	0/8	0/2	0/2	0/0	0/0
0.05	0/14	0/2	0/2	0/0	0/0
0.1	4/26	2/8	3/9	4/7	1/4
0.15	5/35	3/14	2/15	2/12	2/9
0.2	4/46	0/14	4/24	5/17	3/10
0.25	0/46	0/14	1/27	0/18	0/11
0.375	0/46	0/14	4/31	2/19	0/11
0.5	0/46	0/14	5/38	3/26	2/17
0.75	0/46	0/14	2/40	3/28	1/17
1	0/46	0/14	2/40	1/28	0/19
2	0/46	0/14	0/40	2/41	1/28
4	0/46	0/14	0/40	0/41	0/31
8	0/46	0/14	0/40	0/41	0/36
16	0/46	0/14	0/40	0/41	0/36

Table 8: Lung: tightened cell–gene questions / questions with marginal width at least two at each budget. Zero over zero denotes no eligible comparison, not a measured zero effect rate. Each question can recur at multiple budgets.

Area (μm^2)	Adaptive	0.5	1	2	3
0	0	0	0	0	0
0.005	1431	915	768	514	395
0.01	1927	1275	1071	742	557
0.025	2807	1944	1594	1124	834
0.05	3684	2640	2165	1556	1183
0.1	4414	3465	2817	2049	1583
0.15	4896	3852	3254	2376	1808
0.2	5226	3852	3563	2607	2002
0.25	5259	3852	3850	2830	2167
0.375	5259	3852	4411	3278	2508
0.5	5259	3852	4770	3545	2711
0.75	5259	3852	4856	3951	3054
1	5259	3852	4856	4225	3258
2	5259	3852	4856	4814	3670
4	5259	3852	4856	4876	3962
8	5259	3852	4856	4876	4013
16	5259	3852	4856	4876	4013

Table 9: Breast format-test: non-singleton intervals on the common 17-budget grid. Column labels after Adaptive are required physical support halfwidths in micrometres. The numbers of question entries for the five families are 22976, 22538, 23013, 23687, 23772.

Area (μm^2)	Adaptive	0.5	1	2	3
0	0/0	0/0	0/0	0/0	0/0
0.005	9/77	5/30	2/17	0/9	0/5
0.01	21/147	7/66	3/42	3/19	1/8
0.025	50/301	26/159	14/107	11/63	1/23
0.05	84/516	39/289	23/209	12/127	7/64
0.1	80/706	76/469	52/346	39/227	23/142
0.15	135/900	89/598	91/492	55/331	29/212
0.2	127/1021	24/598	82/579	46/383	29/249
0.25	61/1026	4/598	110/674	74/452	47/291
0.375	14/1026	1/598	147/833	91/564	51/371
0.5	3/1026	0/598	148/959	91/651	56/430
0.75	0/1026	0/598	52/990	124/780	85/528
1	0/1026	0/598	8/990	125/873	86/605
2	0/1026	0/598	0/990	121/1102	83/773
4	0/1026	0/598	0/990	8/1134	67/878
8	0/1026	0/598	0/990	0/1134	2/900
16	0/1026	0/598	0/990	0/1134	0/900

Table 10: Breast format-test: tightened cell–gene questions / questions with marginal width at least two at each budget. Zero over zero denotes no eligible comparison, not a measured zero effect rate. Each question can recur at multiple budgets.

8 Reference segmentations and consistent reach denominators

8.1 Image-only nuclear reference

The image-only reference assigns each retained RNA to the nearest labelled nuclear pixel within five physical micrometres, or to background. Nuclear pixels carry their source cell-mask label. RNA coordinates stay fixed. A five-micrometre crop-edge guard excludes points whose nearest nucleus could lie outside the observed crop. This image-only reference provides a deterministic comparison with the expression-informed Proseg output.

8.2 Proseg executions

Proseg 3.2.0 was run on the two lung crops. The initial default-factorization run on crop 1 failed during feature selection with an empty-input error. One completed stochastic run per crop used factorization disabled; remaining sampler settings were 200 burn-in iterations, 200 sampling iterations and 50 hill-climb steps, with two requested threads. Diffusion and source reassignment remained enabled. This version exposes no seed argument. Commands, input/output hashes, exclusions and the initial failed log are preserved alongside the raw results. The same two Proseg outputs are compared with every width family.

Proseg execution quantity	Lung crop 1	Lung crop 2
Input RNA	5,847	1,523
Output RNA	5,774	1,482
Excluded RNA	73	41
Changed source owner	3,242	765
Output background/unassigned	2,484	1,335
Verified nonzero output-count entries	1,443	67

Table 11: Independent ingestion and count-matrix reconstruction of the Proseg executions. Background assignments account for 43.0% and 90.1% of output transcripts. The comparison therefore measures the reach of local boundary deformations against substantial assignment changes on clipped crops.

The check joins exact transcript identifiers, verifies gene names and source assignments, maps Proseg cell labels using the original-cell mapping, and treats the output background flag explicitly. Observed-coordinate serialization residuals are at most 0.000243 and 0.000485 micrometres, respectively. Every saved sparse count entry is reconstructed exactly from the returned assignments. This arithmetic check is independent of the original comparison script; source data and Proseg executions are shared.

8.3 Countwise reach and assignment-vector exclusion

All five width families use a common set of comparison pairs per reference. Retained transcripts are intersected by exact identifier. Pairs whose source count involves an excluded transcript are removed. If a missing transcript is editable in any compared atlas, its gene is also excluded for every candidate destination cell in that patch. These exclusions are applied to every width; all remaining reference and source counts use the same retained observations. Thus denominators remain fixed across widths, and missing Proseg records or crop-edge nuclear exclusions cannot masquerade as changes.

The common comparison has 2,798 changed lung Proseg counts, 1,114 changed lung nuclear counts and 10,965 changed breast nuclear counts. These differ from the original adaptive-only nuclear comparison (1,128 and 11,151 changed pairs) because the new exclusion guard covers the union of editable observations over every width. Included and excluded pairs and reasons are supplied explicitly.

At each budget a reference count is attainable exactly when its count-specific minimum area is within budget. Unbounded-budget reach tests membership in the entire finite count spectrum. Tables 6 and 2 give the latter results; all 17 finite budgets are exported. The complete reference assignment vector is ruled out in each crop at every width by at least one changed fixed RNA or unavailable destination. This exclusion is

stronger than merely observing that some count misses its interval. Conversely, countwise inclusion cannot establish a jointly realizable reference matrix.

A widening ribbon need not improve aggregate reach because its required support removes other patches. The width study separates the displacement available inside accepted ribbons from the extent of tissue covered by those ribbons.

8.4 Co-detection in the primary atlas

The three tested pairs are EPCAM/PTPRC, EPCAM/CD3D and EPCAM/PECAM1. The primary atlas contains 150 lung and 1,148 breast cell/pair questions, each evaluated at eight budgets. Table 12 separates persistent co-detection from source positives that can be removed and source negatives that can become positive. The classifications reached by area $0.25\,\mu\mathrm{m}^2$ persist through the largest tested area of $4\,\mu\mathrm{m}^2$. The vector spectra and marginal intervals give the same classifications on these inputs.

	Pair	Source positive	Removable	Persistent	Newly possible
Lung	EPCAM/PTPRC	7	2	5	0
	EPCAM/CD3D	3	0	3	0
	EPCAM/PECAM1	3	0	3	0
Breast	EPCAM/PTPRC	48	2	46	4
	EPCAM/CD3D	0	0	0	0
	EPCAM/PECAM1	47	5	42	3

Table 12: Co-detection at area $0.25\,\mu\mathrm{m}^2$ in the primary atlas. Removable and persistent counts partition the source positives; newly possible counts describe source-negative cells. Each classification concerns joint positivity of the indicated pair.

Interpretation. These experiments quantify sensitivity within the stated geometric family. Fixed seams and junctions restrict its reach, and its width and budget parameters have yet to be anatomically calibrated. The reference comparisons use two crops per tissue; repeated budget queries provide computational replication rather than an independent donor cohort. The breast crop provenance limits biological interpretation. The nuclear and Proseg comparisons measure alternative assignments; they provide neither transcript-level truth nor a segmentation-accuracy benchmark. Likewise, stable or removable co-detection alone does not identify a cell state, doublet or leakage.

9 Validation and resource measurements

Verification separates the optimal count from the boundary that realizes it. An independent scalar implementation uses exact rational arithmetic to recompute every target assignment's minimum area, then combines local states by Cartesian products. It imports none of the production implication, envelope, integrated-layer, min-cut or count-chain routines, although both implementations use the same prepared inputs. A further direct calculation verifies each witness's slope, band, owners, nuclear constraints and area.

The count-chain test suite additionally compares 201 scalar spectra on 100 generated systems, tests 80 nonflat models against 1,644 exhaustive assignments, verifies positive mandatory deformation and rejection of free interval-valued RNA, and checks the explicit eight-unit gap. All 944 multicopy local gene questions from both data collections agree with independently enumerated spectra. A full hybrid reanalysis uses the chain for all multicopy genes and closure for single-copy genes; every global minimum area and every interval CSV row matches the separately checked enumerated calculation. The transversely monotone gluing map passes inverse, trace, chart roundtrip and independent polygon-area checks.

Check	Lung	Breast format-test crops
Independent local gene spectra	706	7,068
Target binary assignments	1,498	17,542
Local witness records	1,101	11,594
Global spectra	3,304	22,976
Cartesian tuples	3,905	32,730
Global attaining count witnesses	3,876	29,799
Unique budget intervals	26,432	183,808
Exact half-ribbon/pixel checks	31,642	105,824
Relevant ribbon-pair checks	491	1,602
Nuclear pixel fragments	6,609	42,732

Table 13: Exact validation of the primary atlas: all comparisons agree and every attaining witness satisfies the continuous constraints.

9.1 Width-study checks

All eight width analyses completed, giving 106,351 global spectra and 1,807,967 interval rows. Local attaining boundaries were checked in continuous coordinates during computation. A separate composition routine reconstructed every global spectrum and checked 131,376 attaining traces, including source counts, local contribution signs, total area and unique transcript support. Every interval row was then recomputed from its exact spectrum and compared with its marginal outer bound.

For an independent optimality comparison, a fixed input-hash rule selects up to 32 questions per setting with two–six target copies. Exhaustive scalar enumeration recovers all 246 spectra from 1,628 target assignments, within the 15-second per-query limit. Forced-owner preprocessing is also tested on 80 generated models: 160 spectra, 773 target assignments and 287 witnesses agree among the original chain, the pruned chain and separate scalar enumeration. Preprocessing substitutes only hard-implication consequences; it changes no feasible owner vector.

A single process per width computes its complete spectra with a 60-second local chain allowance. Timings for these simultaneous runs record their computational cost; the sequential repeated scaling experiment below measures runtime under a controlled schedule. All computations use the same ingested observations, masks and prepared atlas geometry.

9.2 Union reconstruction and witness checks

A separate implementation checks the common source hashes, retained RNA records, complete source counts and positive-or-editable question census across all members. It independently merges the validated component spectra and recomputes all 29,212 union spectra, 42,123 finite count states and 496,604 interval rows. It also checks all 2,254,727 member rows on the 17-budget grid, all 68 population/budget summaries and the three reference-reach totals.

Each union count state carries one member identity. The checker validates 39,191 explicit whole-atlas traces and 2,932 fixed-count completions, including 18,762 distinct local boundary witnesses. Direct piecewise-linear checks verify slopes, bands, clearance, nuclear pins, endpoint seams and integrated rational area. Five deliberately invalid boundary mutations are rejected. All checks pass in 142.29 seconds in the recorded environment. The component minimum-area values are those validated in the primary and width analyses above; this new pass independently checks their union and attaining realizations.

9.3 Count-chain scaling

The fixed constructed family has flat source, slope one, equal flipped height $h = 1/4$, site spacing $d = 1/10$, clearance $1/1000$, a containing band and endpoints outside all tents. The minimum for $k \geq 1$ changes is

$$A_k = h^2 + (k-1)(hd - d^2/4), \qquad A_0 = 0,$$

attained by k consecutive sites. For arbitrary selected sites, each additional separation contributes at least the adjacent-site area increment, which proves the lower bound. The closed-form spectrum provides an exact reference for measuring the general count chain as input size grows.

RNA	Count levels	Mesh nodes	Height candidates	Visited states	Median total(s)
16	17	111	1,154	8,112	0.465
32	33	207	2,370	35,104	1.574
64	65	399	4,802	147,456	7.138
128	129	783	9,666	605,632	42.206

Table 14: Three fresh sequential processes per size. Total includes preparation, dynamic programming, all witness reconstruction and continuous checks. No limit is reached.

Figure 4 shows the three measurements at each size. At 128 RNAs, total runtime ranges from 40.2 to 42.7 seconds, with median mesh preparation of 2.0 seconds and dynamic programming plus witness reconstruction and verification of 40.2 seconds. Separate medians need not sum exactly. The tested limits are 300,000 retained candidate heights, one million visited states, 60 seconds for preparation, 90 seconds for dynamic programming, and 160 seconds for a complete process. Requested single-thread environment variables are recorded, but the arithmetic is Python integer/Fraction work. Peak whole-process RSS at the largest size is approximately 205 MiB and includes imports. Raw process records retain the full timing precision and environment.

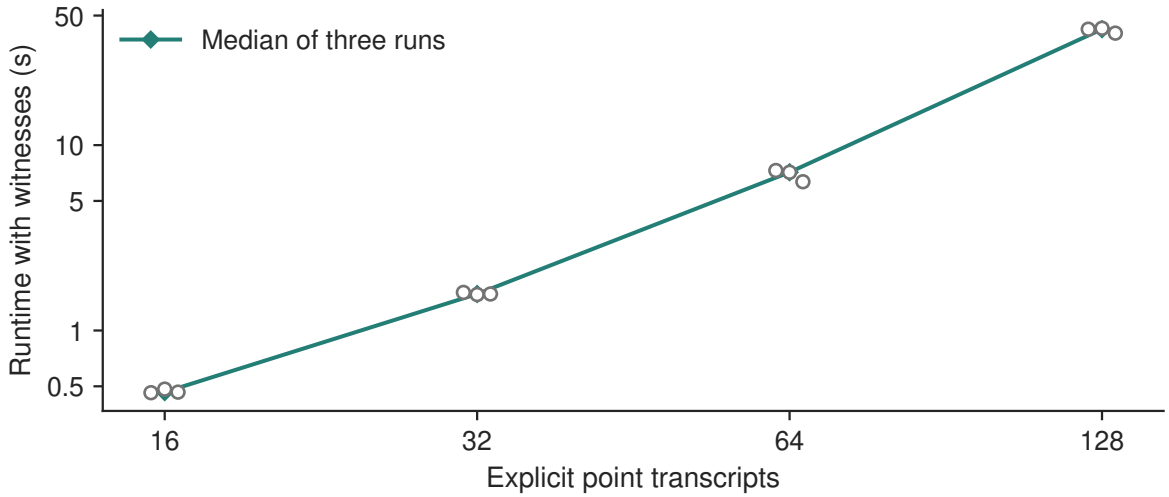


Figure 4: Count-chain runtime on the constructed family with known exact optima. Open circles show three fresh processes per size, offset horizontally for visibility; diamonds connect their medians. Runtime includes preparation, all counts, witness reconstruction and verification. Both axes are logarithmic. These measurements characterize this family rather than predict runtime for arbitrary tissue ribbons.

9.4 Contiguous native-mask regions

We acquired the complete breast format-test source and fixed central squares with nominal sides 400 and 600 μm from the raster dimensions before extracting their transcripts or computing counts (Figure 5). The 400 μm selection uses pixel bounds $[1955, 3837) \times [823, 2705)$, giving a 1882×1882-pixel region with actual side approximately 399.925 μm. The 600 μm selection uses $[1484, 4307) \times [353, 3176)$, giving 2823×2823 pixels and an actual side of 599.888 μm. These extend the contiguous computational footprint within the same format-test source. The regions overlap previous crops, so their cell totals are reported separately.

An independent source audit imports no preparation adapter or GeoCert module. It verifies the archive and seven member hashes, rational pixel assignments for all 1,035,543 Q20 measured-gene source candidates, and both regions' prepared masks, retained RNA records and source counts. Both inputs pass, with zero source-owner disagreements.

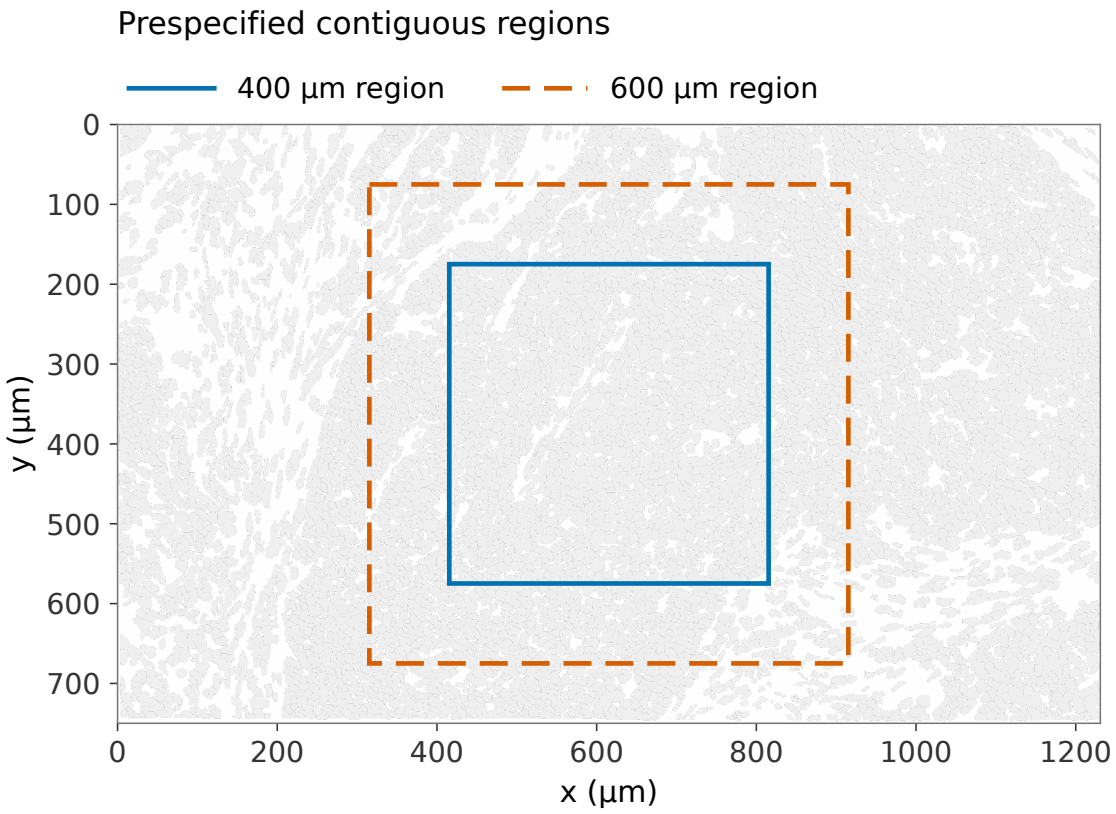


Figure 5: Prespecified contiguous regions in the original breast format-test raster. Central squares with nominal sides 400 μm (blue, solid) and 600 μm (orange, dashed) were fixed from raster dimensions before transcript extraction. Pixel-aligned sides are approximately 399.925 and 599.888 μm; the source extent is 1230.800×749.913 μm. Outlines identify the selected regions.

The completed 400 μm analysis contains 1,363 crop-contained connected cells and 226,983 retained Q20 measured-gene transcripts. The primary adaptive extraction rule produces 5,102 disjoint ribbons, containing 29,011 transcripts and covering 137,516 of 211,770 eligible incident edges (64.94%). The unchanged chain backend computes 69,278 cell–gene spectra and 554,224 exact rows on the original eight-budget grid. The largest ribbon contains 50 RNAs and the largest local gene query has seven copies; 3,594 chain queries visit 7,039,980 height/count states. This region evaluates the primary family, separately from the five-family union above. There are no failed geometry checks or resource-limited local queries.

The completed 600 μm analysis contains 3,171 complete cells, 470,293 retained transcripts and 11,748 primary-atlas ribbons. These contain 62,108 transcripts and cover 318,946 of 478,423 eligible incident edges

(66.67%). The exact enumeration backend computes 152,925 cell–gene spectra and 1,223,400 rows on the same eight-budget grid. A backend amendment recorded before this count computation permits enumeration only after an input check establishes at most 16 target copies per local gene, within a 65,536-pattern cap. The largest observed query has seven copies. The completed result has zero failed geometry checks and zero unresolved or resource-limited local queries.

	400 μm: chain		600 μm: enumeration	
Stage	Wall (s)	RSS (MiB)	Wall (s)	RSS (MiB)
Atlas extraction from masks	152.46	960.43	423.93	1,920.03
Rational geometry validation	199.66	364.88	480.19	780.73
Counts, witnesses and serialization	1,000.91	2,488.77	878.95	2,251.76
Completed stages	1,353.03	—	1,783.07	—

Table 15: Completed stages on the two prespecified regions. Wall times include startup and serialization; each stage runs in a fresh process. RSS is peak process memory including imports. The 600 μm column excludes an interrupted count attempt; its cumulative attempt cost is reported in the text. The regions use different exact backends.

The 600 μm run initially exhausted its whole-run allowance during enumeration, after 1,321.07 seconds. A recorded amendment then allowed one retry of count optimization and witness generation, capped at 1,200 seconds, using the same validated atlas, backend and local limits. The completed retry takes 892.04 seconds including its outer checks. Cumulative measured attempt time is therefore 2,213.11 seconds (36.89 minutes), including the interrupted work. This differs from the 1,783.07-second sum of successfully completed stages in Table 15. The failed attempt and amendment remain in the record. Source-crop preparation takes 12.44 and 20.74 seconds for the two regions; coverage accounting takes 1.02 and 2.38 seconds, respectively, separately from the completed-stage sums.

Independent output audits take 19.65 and 33.69 seconds, separately from these run times. For the 400 μm region, the audit checks all 24,132 local spectra, 39,519 local witness records, 95,253 cell-state traces and 554,224 interval rows. For the 600 μm region, it checks 52,169 local spectra, 85,335 local witness records, 208,555 cell-state traces and all 1,223,400 rows. The audits independently reconstruct min-plus compositions and reconcile recorded witness ownership counts and areas. Local optimality is provided by the unchanged exact solvers, which also check continuous boundaries.

Separate exact rational scalar enumeration checks 16 local multicopy queries per region, using a deterministic input-only hash rule and importing no production geometry or optimization routines. The 400 μm selection was fixed before production from 3,590 eligible queries with two–six target copies; the selected queries have two–four copies. The same pre-existing rule selects 16 of 7,438 eligible 600 μm queries after that production run, before inspecting its spectra; selected queries have two–six copies. The checks enumerate 92 and 152 target assignments and recover 38 and 33 finite count states, respectively. All 32 complete spectra match exactly, with no errors or resource limits under a 15-second per-query cap.

The run uses Linux and Python 3.12.14 with numerical-library thread limits set to one, while unrelated work runs in the shared environment. The timings describe this workload under those conditions. The source archive hash, selection protocol, stage logs and output hashes accompany the results.

9.5 Source feasibility and complete-atlas composition

The general count chain allows a geometrically feasible reference whose original molecular assignments require positive area. The tissue adapter instead requires zero-cost source assignments before using implication closure or omitting patches from a count query. In the original analysis, patches containing RNA pass that guard; RNA-free patches are skipped by the count loop. We therefore checked the source condition directly on every saved patch, including the skipped ones.

An independent rational calculation substitutes $h(t) = 0$ into all RNA-clearance and nuclear-ownership inequalities and verifies the band, slope and fixed-endpoint constraints. It imports no compiler, optimizer, closure or witness routine. Across the primary and fixed-width studies, two biological donors and the contiguous regions, all 28 distinct atlases pass: 26,349 patches, including 4,793 RNA-free patches. The archive inventory contains 30 records because the two lung primary atlases also appear in the external-validation package. Every omitted patch therefore has minimum required area zero, so the corrected composition leaves all reported spectra unchanged. This verifies a premise on the saved scalar models; the earlier source-data and geometry audits establish their relation to the tissue inputs.

The complete-atlas implementation requires a local spectrum for every declared patch, retains its cost at change zero, and returns a complete tuple of attaining witnesses. It rejects missing or unfinished spectra; an empty local feasible set makes the global set empty. It uses the existing min-plus routine, which already sums all supplied local costs.

A regression makes the positive-cost case explicit. Consider two disjoint scalar charts with source zero on $[0, 4]$, slope one, band $[-1, 1]$ and clearance $1/4$. A target RNA at $(t, x) = (2, -1/2)$ has source owner zero. Keeping it costs zero; transferring it requires a downward tent of height $3/4$ and area $9/16$. The second chart has a non-target RNA at $(2, 0)$: either owner requires a tent of height $1/4$, so its only count-change state is zero with area $1/16$. The complete source count is one, and the exact global minima are

$$S(1) = 1/16, \qquad S(0) = 9/16 + 1/16 = 5/8.$$

At budget $3/5$, the complete interval is $[1, 1]$, whereas dropping the non-target patch would incorrectly yield $[0, 1]$. The unchanged critical-height solver reproduces these analytic areas, and direct continuous checks verify both local boundaries in each global witness. Further cases cover an infeasible non-target patch, a fixed count with positive mandatory area, a zero-cost source, and rejection of incomplete inputs. These tests concern scalar composition; they do not provide an end-to-end test of the general coupled-interface construction.

The scripts and their recorded outputs are in `extensions/composition_audit/`. The source-feasibility audit reads the archived atlases, while the regression constructs the small systems above; neither changes the tissue results.

10 Replication on biological breast tissue and segmentation context

10.1 Data and selection

We used the public 10x FFPE Human Breast with Pre-designed Panel dataset, processed with Xenium Onboard Analysis 1.0.2 [4]. It contains an infiltrating ductal carcinoma sample with ductal carcinoma in situ and an invasive lobular carcinoma sample from two donors. Both are distinct tissue sources from the lung donor. The donor identity behind the earlier breast format-test input is unspecified, so we make no claim that it supplies an additional independent breast donor. The new sources use DAPI nuclear segmentation and vendor expansion by 15 μm; their native cell masks define the reference partitions.

For each donor, we formed three target locations using the 25th, 50th and 75th percentiles of cell-centroid x and the median cell-centroid y. The nearest cell centroid to each target fixed a nominal $200 \times 200\,\mu$m core and an $800 \times 800\,\mu$m context window. Origins were rounded down to the native pixel grid. The selection used no gene identities, transcript counts or GeoCert outcomes. All six cores were retained. Their coordinates, source-member checksums and selection rules are recorded with the analysis.

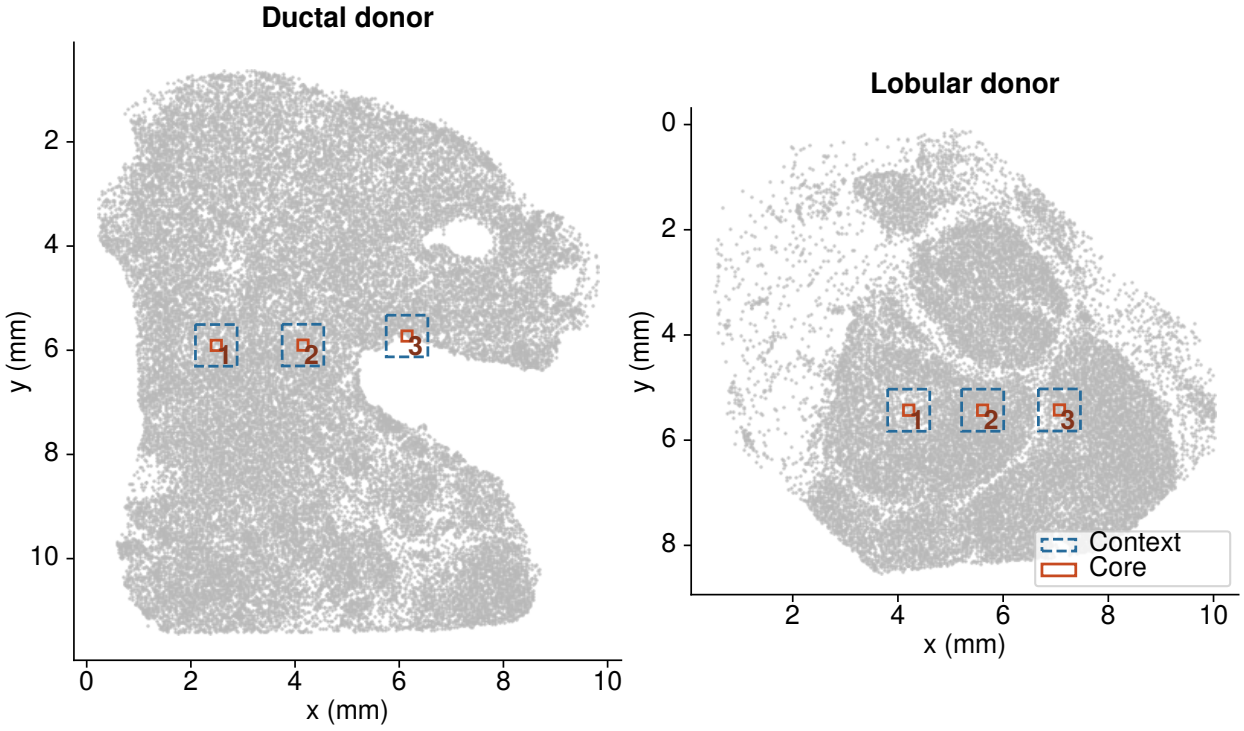


Figure 6: The six predefined breast tissue regions. Grey points show a regularly subsampled set of source cell centroids; solid boxes mark the 200 μm analysis cores and dashed boxes the 800 μm Proseg inputs. Each panel represents one donor.

The original cell and nucleus raster masks, physical coordinate transform and transcript identifiers were retained. We included every Q20 transcript targeting a measured gene in the selected regions. An independent pass through the original Parquet files reproduced all retained identifiers, coordinates, raster assignments and source counts. There were zero discrepancies between retained transcript owners and the native masks. Cells touching a core edge or lacking a connected valid reference region were excluded by the unchanged atlas eligibility rule.

10.2 Exact count intervals and matched approximations

The primary atlas uses the original ribbon-width ladder, slope, nuclear constraints and fixed seams, with the original eight area budgets. Questions cover positive or editable cell–gene pairs, retaining the fixed counts outside ribbons. Extraction and independent rational checks precede count computation. We used exact implication-pruned enumeration for local single-gene spectra; every spectrum completed. Separate scalar calculations reproduced all local minima, and Cartesian composition and explicit boundary checks verified every global spectrum and endpoint. The analysis therefore tests the existing model on new biological samples without changing its geometric assumptions.

Quantity	Ductal donor	Lobular donor
Complete eligible cells	659	859
Retained core transcripts	97,893	91,659
Boundary ribbons	1,531	1,946
Covered eligible shared edges	76.5%	76.5%
Cell–gene questions	32,360	36,409
Invariant at $4\,\mu m^2$	25,749 (79.6%)	30,403 (83.5%)
Questions with outer width at least two	1,271	829
Questions tightened by shared geometry	176 (13.8%)	97 (11.7%)
Questions with an additive missed endpoint	58	37

Table 16: Results across three predefined cores per donor. Invariance is conditional on the primary atlas's approximately $0.45\,\mu m$ displacement cap and fixed uncovered boundaries. Eligibility, tightening and additive misses are counted once per cell–gene question across the eight budgets. Tightening affects 0.54% and 0.27% of all questions, respectively.

Shared geometry tightens the independent-molecule bound in every selected region (Table 17). The pooled donor fractions sum tightened and eligible pairs across regions; they are not averages of regional percentages. Eligibility is fixed over the full eight-budget grid, so the same pairs form the denominator for each budget in the main paper's donor plot.

Region	Complete cells	Cell–gene pairs	Eligible pairs	Tightened pairs
Ductal 1	208	9,712	243	31 (12.8%)
Ductal 2	300	14,571	642	84 (13.1%)
Ductal 3	151	8,077	386	61 (15.8%)
Lobular 1	219	9,204	186	15 (8.1%)
Lobular 2	358	15,233	481	63 (13.1%)
Lobular 3	282	11,972	162	19 (11.7%)

Table 17: Regional breakdown of Table 16. Eligibility requires an independent-molecule interval width of at least two at some tested budget; tightening is counted once per pair across budgets. The three regions within each donor are disjoint tissue samples, not independent donors.

The independent-molecule outer bound uses the same exact singleton transfer costs as the original analysis. The additive inner approximation retains assignment feasibility while replacing shared area by the sum of singleton costs, including collateral transfers. These comparisons separate the effects of shared geometry and shared deformation cost. A further distance-only outer bound counts molecules near raster boundary edges using the global displacement cap and a half-pixel allowance for representing each unit edge by its midpoint. It relaxes the ribbon, nuclear and area constraints; tightening against that looser bound alone is not evidence of a shared-cost effect.

Budget (μm^2)	Ductal donor		Lobular donor	
	Non-singleton	Tighter outer	Non-singleton	Tighter outer
0	0	0	0	0
0.05	5,082	94	4,551	40
0.1	5,770	62	5,249	43
0.25	6,611	38	6,006	17
0.5	6,611	0	6,006	0
1	6,611	0	6,006	0
2	6,611	0	6,006	0
4	6,611	0	6,006	0

Table 18: Budget-specific counts in the two biological donors. The shared-geometry differences occur at smaller budgets; the singleton outer bounds coincide with the exact intervals from $0.5\,\mu m^2$ onward on this grid.

10.3 Paired Proseg context analysis

For each breast core and both original lung crops, we ran Proseg 3.2.0 on the core alone and with $300\,\mu m$ of surrounding tissue on each side. Default factorization and the earlier no-factorization setting were tested separately, keeping 200 burn-in iterations, 200 sampling iterations, 50 hill-climbing iterations and two threads. The default run on the first small lung crop again failed; it was retained as a failed setting rather than replaced. Each completed setting is one stochastic run, so differences are descriptive and include sampling variability.

Comparisons use the same original core transcript identifiers retained by all completed settings for that core, with output cell identities joined to source identities. Background assignments override the reported cell index. We reconcile final transcript assignments from the native Proseg Parquet output with its sparse count matrix; optional CSV exports are not required for this check. All settings for a core use the union of their source and alternative cell–gene questions. Queries affected by an excluded editable transcript are marked unavailable rather than scored with a shifted full-transcript spectrum. We report cell-to-cell transfers and background transfers separately. Countwise inclusion in an interval is distinguished from realizability of the complete assignment vector.

The larger input window reduces the background fraction in both lung cores and all six breast cores under the no-factorization setting (Figure 7). This comparison holds core transcript identifiers fixed: the plotted change reflects the additional surrounding input, together with stochastic variation, rather than a change in the evaluated tissue region. The full setting tables below also retain default-factorization runs and the failed run.

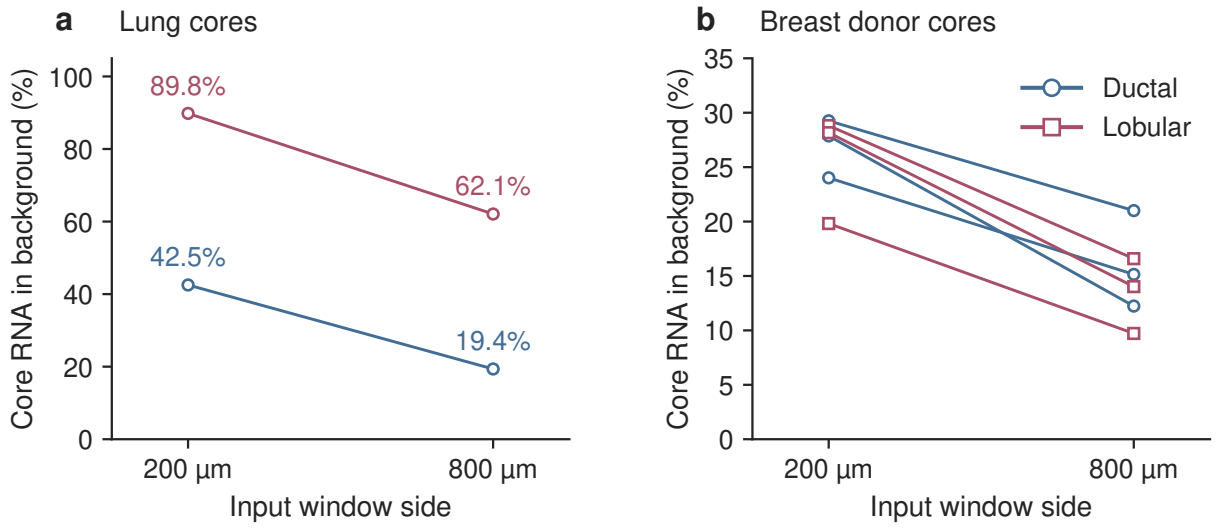


Figure 7: Proseg background assignments for the same core molecules with different surrounding input. Each line connects one core's 200 and 800 μm no-factorization runs: (a) the two lung crops and (b) three cores per breast donor. Each point is one stochastic execution, without replicate-based uncertainty estimates. Vertical scales differ between panels.

Core	Retained/input RNA	Default factorization		No factorization	
		200 μm	800 μm	200 μm	800 μm
LUNG 1	5,774/5,847	failed	19.5	42.5	19.4
LUNG 2	1,482/1,523	86.5	60.7	89.8	62.1
IDC 1	27,250/27,250	29.2	21.3	29.3	21.0
IDC 2	45,198/45,198	24.2	14.8	24.0	15.1
IDC 3	25,329/25,445	27.9	12.5	27.9	12.2
ILC 1	22,791/22,791	27.7	16.6	28.8	16.6
ILC 2	41,168/41,168	20.1	9.8	19.8	9.7
ILC 3	27,700/27,700	27.1	13.8	28.2	14.0

Table 19: Percentage of retained core transcripts assigned to background. Each row uses the intersection of core transcript identifiers across completed settings; the second column reports its size against the original core input. Input width describes the surrounding window, while evaluation always uses the central core.

Core	Setting	Changed counts	Reach (%)	With BG (%)	Fixed RNA
LUNG 1	200 / D		Factorization failed		
LUNG 1	200 / NF	2,262	7.2	68.0	2,777
LUNG 1	800 / D	2,198	8.2	39.4	2,292
LUNG 1	800 / NF	2,139	7.8	40.0	2,347
LUNG 2	200 / D	513	5.1	90.4	683
LUNG 2	200 / NF	504	5.2	95.4	684
LUNG 2	800 / D	433	5.3	75.1	519
LUNG 2	800 / NF	470	4.9	75.3	571
IDC 1	200 / D	7,113	7.1	60.3	11,280
IDC 1	200 / NF	7,048	6.9	61.3	11,226
IDC 1	800 / D	6,496	7.9	45.7	9,572
IDC 1	800 / NF	6,541	7.9	45.3	9,473
IDC 2	200 / D	11,107	7.9	57.0	17,495
IDC 2	200 / NF	11,094	7.8	56.4	17,549
IDC 2	800 / D	10,059	9.2	37.0	14,560
IDC 2	800 / NF	10,126	9.1	38.1	14,597
IDC 3	200 / D	6,226	8.4	66.1	10,321
IDC 3	200 / NF	6,363	7.8	66.0	10,286
IDC 3	800 / D	5,660	9.8	33.8	7,794
IDC 3	800 / NF	5,570	10.3	34.1	7,689
ILC 1	200 / D	6,558	6.8	62.3	8,951
ILC 1	200 / NF	6,662	6.6	62.9	9,073
ILC 1	800 / D	5,729	8.0	41.1	6,897
ILC 1	800 / NF	5,801	8.0	41.3	6,862
ILC 2	200 / D	10,975	8.2	51.8	14,360
ILC 2	200 / NF	11,021	8.5	50.9	14,446
ILC 2	800 / D	9,699	9.6	27.7	11,234
ILC 2	800 / NF	9,651	9.9	26.9	11,202
ILC 3	200 / D	8,166	6.1	61.3	10,366
ILC 3	200 / NF	8,240	5.9	62.6	10,626
ILC 3	800 / D	7,227	7.2	35.5	7,768
ILC 3	800 / NF	7,297	7.0	35.7	7,838

Table 20: All paired Proseg settings. D: default factorization; NF: no factorization. Reach is the fraction of changed cell–gene counts individually attainable within the primary atlas at $4\,\mu\mathrm{m}^2$. With BG counts changed entries touched by at least one transfer to or from background; these may also involve cell-to-cell transfers. Fixed RNA moves count reassigned transcripts outside editable ribbons. Every completed full assignment lies outside the family. Each core uses a fixed set of cell–gene questions across its completed settings.

In the original lung cores, the matched no-factorization background fractions change from 42.5% to 19.4% and from 89.8% to 62.1% when surrounding tissue is included. Thus the extreme background rates in the small-crop comparison depend on the input context. Larger inputs still produce many changes outside the primary family (Table 20). These results diagnose the comparison setting; they do not establish which assignments are anatomically correct or isolate context effects from stochastic variability.

The public artifacts retain core transcript assignments for every completed run, the complete setting inventory, countwise comparisons and audit results. Replaying those extracts reproduces the tables; repeating Proseg itself is stochastic. Native transcript outputs were reconciled with the full output count matrices before the core extracts were saved.

11 Reproduction, data rights and assistance

The prepared-input exact workflow uses Python's standard library. Geometry extraction, raster reference analysis and plotting additionally use the declared analysis dependencies. The main commands are:

```
python atlas_cli.py verify-manifest
python release/restore_results.py
python atlas_cli.py showcase --output run/showcase.json
python atlas_cli.py validate
python analysis/reach/audit_proseg.py
python analysis/reach/check_outputs.py
python analysis/reach/report.py
```

The README gives commands for the tables, figures and full count analyses at each width. Restoring compressed results verifies both stored and expanded checksums. The archive includes spectra, witnesses, comparison pairs, Proseg records, selection rules, reference masks, numerical checks and runtime environments.

GeoCert is MIT-licensed. The 10x data retain CC BY 4.0 attribution and the breast format-test limitation; Proseg retains its upstream rights, and its executable is not redistributed. Code, inputs and exact results are archived at `https://github.com/arnavperam/geocert-recomb2027`. Archive and document checksums are recorded in the repository.

Generative AI was used solely for software implementation, testing, and drafting.

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